PHY 115L
Variational Methods

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1 TISE and System of Units

We will use a variational approach to find the ground state $\Psi_0(x)$ of the Time-Independent Schrödinger Equation (TISE):

$$\hat{H}\ \psi_0(x) = E_0\ \psi_0(x)$$

where the Hamiltonian operator is defined by:

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x) \tag{1}$$

As before, we will use a system of units based on a characteristic length scale a and an energy:

$$\epsilon \equiv \frac{\hbar^2}{2ma^2} \tag{2}$$

(In previous homeworks we called this quantity E_0 but this could too easily be confused with the ground state!) With these units, we have:

$$\frac{\hat{H}}{\epsilon} = -a^2\frac{d^2}{dx^2} + \frac{V(x)}{\epsilon} \tag{3}$$

2 Numerical Method for Finding the Second Derivative

We will calculate second derivatives using a second-order approximation:

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

We'll assume that we calculate our function with a constant spacing so that:

$$x_j = x_0 + h * j$$

and given y values:

$$y_j = f(x_j)$$

we can calculate the second derivative as:

$$f(x_j) = \frac{y_{j+1} - 2y_j + y_{j-1}}{h^2}$$

One challenge with this technique is how to handle the edges. For these exercises, it turns out that simplest solution is just to arrange things so that the second derivative is always zero. This is the case if the edges are at $x = \pm\infty$ (effectively) or if we are considering an *odd* function from $x = 0$ to $x = +\infty$. In lecture, I mentioned that we could use periodic boundary conditions, but I have found the method proposed here is a simpler solution that works in every exercise below.

There are many ways to implement this calculation with numpy array y, but I found the easiest was to use the `np.roll` function, followed by setting the edges to zero by hand.

3 The Variational Method of Finding the Ground State

Our technique is based on the observation that:

$$\langle\psi|\hat{H}|\psi\rangle \geq E_0$$

for all ψ. Which you can see by expanding $|\psi\rangle$ in the complete set of energy eigenstates:

$$|\psi\rangle = \sum c_n \, |E_n\rangle$$

so:

$$\langle\psi|\hat{H}|\psi\rangle = \sum |c_n|^2 E_n = |c_0|^2 E_0 + \sum_{n>0} |c_n|^2 E_n$$

but:

$$\sum |c_n|^2 = 1 \quad \Longrightarrow \quad |c_0|^2 = 1 - \sum_{n>0} |c_n|^2$$

so:

$$\langle\psi|\hat{H}|\psi\rangle = E_0 + \sum_{n>0} |c_n|^2 (E_n - E_0)$$

but as $E_n \geq E_0$ we have:

$$\langle\psi|\hat{H}|\psi\rangle \geq E_0$$

- Choose the range $[a, b]$ of x wide enough so that the wave-function can be considered to be zero outside of this range, with the second derivative 0 at $x = a$ and $x = b$.
- Divide $[a, b]$ into n equal intervals ($n + 1$ discrete points) with

  $$x_j = a + h\,j$$

  for $0 \leq j \leq n$ and step size $h = (b - a)/n$. (This is a good case for `np.linspace`)
- Choose a starting wave function $\psi_I(x)$ (an initial guess).
- Construct an array of y values from: $y_j = \psi_I(x_j)$
- Set $y_0 = y_n = 0$
- Normalize the function represented by y_i, by finding the appropriate scale factor such that:

  $$h \sum_j y_j^2 = 1$$

Next we iterate as needed, doing the following at each iteration:

- Choose a random integer n from 1 to $n-1$ inclusive (notice this avoids the end points, which we are fixing to zero).
- Choose a random (uniform) value δy from $-\Delta y$ to Δy, where Δy is a parameter indicating the maximum amount of change in each iteration.
- Put $y_i \to y_i + \delta y$
- Normalize the y values as above
- Calculate the expectation value for the energy $\langle y|\hat{H}|y\rangle$.
- If the expectation value for the energy has decreased, keep the changes to wave function. If not, discard the change from this iteration, and revert back to y values at the start of this iteration.

It is often helpful to use a process called simulated annealing, where we initial use a large value of Δy, so that the wave function changes rapidly, then reduce the value of Δy for fine tuning.

4 Homework Problems

Problem 1: Implement a function that, when provided with an array of y values, calculates the second derivative using a second-order numerical technique as described above. Validate your function for $f(x) = \sin(x)$ in the range $[0, 2\pi]$ by plotting $f(x)$ and $f''(x)$ as calculated with your code.

Problem 2: Implement a function `norm(x,y)` that takes two arrays x and y of the same size, and normalizes y as described in the write-up. Check your function with code like this:

```
# normalization check:
x = np.linspace(-10,10,100)
y = np.random.uniform(np.size(x))
y=norm(x,y)
print(np.sum(y*y) * (x[1]-x[0]))
```

which should print a number close to 1.

Problem 3: Implement a function `energy(x,y,V)` that returns the expectation value of the energy:

$$\langle\psi|\hat{H}|\psi\rangle$$

The parameters x and y are arrays containing the discrete x-values (x_j) and the discrete y-values ($\psi(x_j)$) for the normalized wave function. The parameter V is a function V(x) that defines the potential. You'll want to use your second derivative function from Problem 1 for the kinetic energy term in the hamiltonian. Test your energy function with code like like this:

```
# free particle energy check:
OFFSET = 10
def V(x):
    return OFFSET
x  = np.linspace(-5,5,1000)
y  = cos(np.pi * x / 2)
y = norm(x,y)
print("calculated energy: ", energy(x,y,V))
print("expected energy:   ", OFFSET+pi**2/4)
```

Your calculated free particle energy should match the expected energy.

Problem 4: Implement the variational method of finding the ground state, as described above. You'll want to use the functions you developed and tested in Problems 1-3. Use it to find the ground state of a (nearly) infinite square well potential:

$$\frac{V(x)}{\epsilon} = \begin{cases} 0, & -a \leq x \leq a \\ B & \text{otherwise} \end{cases}$$

where B is some big number, like $B = 1000$. Plot your initial wave function, your final wave function, and at least one intermediate wave function in the same plot. Estimate your numerical uncertainty. Compare your results to the expected ground state wave function (inside the well):

$$\psi_0(x) = \sqrt{\frac{1}{a}} \cos\left(\frac{\pi x}{2a}\right)$$

with ground state energy:

$$E_0 = \frac{\pi^2 \hbar^2}{2m(2a)^2} = \epsilon\, \frac{\pi^2}{4a}$$

Remember that we set $\epsilon = 1$ and $a = 1$ in the code. There are lots of possible settings, but here are the ones that I used (you can certainly play around with these, if you like, and see how different choices effect convergence.)

- I took:

  $$\psi_I(x) = \begin{cases} 1, & -0.8 \leq x/a \leq 0.8 \\ 0 & \text{otherwise} \end{cases}$$

  as my starting wave function.

- I used 50 bins with x/a in the range $[-1.2, 1.2]$.

- I used simulated annealing: (A) 10,000 iterations with $\Delta y = 0.1$, followed by (B) 10,000 iterations with $\Delta y = 0.01$, then (C) an additional 10,000 iterations with $\Delta y = 0.01$. With these settings I reached 1% accuracy quite rapidly. I estimated this accuracy by the difference in energy after (B) and (C).

Problem 5: Using the variational method of Problem 2, find the ground state of a (nearly) infinite square well potential with a curved bottom:

$$\frac{V(x)}{\epsilon} = \begin{cases} V_p(x) & -a \leq x \leq a \\ B & \text{otherwise} \end{cases}$$

where

$$V_p(x) = -\epsilon \cos\left(\frac{\pi x}{2a}\right)$$

Calculate the change in energy ΔE relative to

$$\frac{E_0}{\epsilon} = \frac{\pi^2}{4a}$$

We can estimate this change to first order using non-degenerate perturbation theory, with

$$\Delta E = \langle \psi_0 | V_p(x) | \psi_0 \rangle$$

This amounts to the integral:

$$\Delta E = \frac{\epsilon}{a} \int_{-a}^{a} \cos^3\left(\frac{\pi x}{2a}\right) = \epsilon \frac{8}{3\pi}$$

Compare your measured energy level shift to the value expected from perturbation theory.

Problem 6: Using the variational method of Problem 2, find the ground state of a harmonic oscillator:

$$V(x) = \epsilon x^2$$

Plot your initial wave function, your final wave function, and at least one intermediate wave function in the same plot. Estimate your numerical uncertainty. Compare your ground state energy to the expected result:

$$E_0 = \epsilon$$

These are the settings I used:

- I used the same initial wave function as in Problem 2.
- I used 50 bins with x/a in the range $[-3, 3]$.
- I used the same simulated annealing as in Problem 2 and reached better than 1% accuracy.